

Chapter 1: Linear Algebra I

Vector Spaces

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Outline

In this chapter, we discuss

- The basic concept of the general, formal vector space concept
- The details of the widely used Euclidean Vector Space
- Mathematical distance functions and their properties
- Limits and continuity beyond univariate real-valued functions
- Key properties of general sets (open/closed, bounded, convex)

What is a Vector Space?

General Vector Space

Collection of objects on which we can perform **linear operations** (addition & scaling)

- Set of vectors V and scalars F with definitions for $v + \tilde{v} \in V$ and $f \cdot v \in V$
- Addition is associative and commutative, scaled addition is distributive
- Additive **identity** ($v + \mathbf{0} = v$) and **inverse** ($v + -v = \mathbf{0}$), scaling identity ($\mathbf{1} \cdot v = v$)
 $\Rightarrow$ general notation $\mathbb{X} = (X, F, +, \cdot)$ [simplify to $\mathbb{X} = (X, +, \cdot)$ for $F = \mathbb{R}$]

Euclidean Vector Space

Resctrict attention to $\mathbb{R}^n$ with **ordered** n -tuple of objects, e.g. $(2, 1, 3)' (\neq (3, 1, 2)')$

- We destinguish row vs. column vector (convention: “vector” = column vector)
- In above language: $F = \mathbb{R}$ and $V = \mathbb{R}^n := \{(x_1, \dots, x_n)' : (\forall i \in \{1, \dots, n\} : x_i \in \mathbb{R})\}$

Linear Combination and Basis

Linear combination (LC) of vectors

- z is a LC of k vectors $x^{(1)}, x^{(2)}, \dots, x^{(k)} \in X$:
 $z = \sum_{i=1}^k \lambda_i x^{(i)}$ with $\lambda_i \in \mathbb{R} \forall i \in \{1, \dots, k\}$

Span = set of linear combinations of $\{x^{(1)}, x^{(2)}, \dots, x^{(k)}\}$, e.g.:

$$\text{Span} \left(\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\} \right) = \left\{ \begin{pmatrix} \lambda + \mu \\ \mu \\ 0 \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : x_3 = 0 \right\}$$

Basis of a vector space: smallest set of vectors to *span* the space

- Basis is not unique: e.g. $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ vs. $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ for $\mathbb{R}^2$
- Canonical basis $\{e^1, \dots, e^n\}$ with $e_i^i = 1$ and $e_j^i = 0$
- **Dimension** of vector space: number of elements in the basis

Linear Dependence and Linear Independence

- Linear dependence: $S \subseteq X$ set, $x \in X$ vector
 x is **linearly dependent** of S if it is a **LC of elements** in S , that is, $x \in \text{Span}(S)$
- Linear independence (LI): $x \notin \text{Span}(S)$ (x is no LC of S)
LI set $B \subseteq X$: every element is LI of the remaining set: $\forall b \in B : (b \notin \text{Span}(B \setminus \{b\}))$
 e.g. $\{e_1, e_2\}$: $\text{Span}(\{e_1\}) = \{(x, 0)' : x \in \mathbb{R}\} \not\ni e_2$ and vice versa

Testing Linear Independence

The set of vectors $B = \{b_1, b_2, \dots, b_k\}$ are linearly independent if

$$\sum_{j=1}^k \lambda_j b_j = \mathbf{0} \Rightarrow (\forall j \in \{1, \dots, k\} : \lambda_j = 0).$$

Example: Linear Independence of the Canonical Basis

The vectors

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \mathbf{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^n$$

are linearly independent, because if $\lambda_1 \mathbf{e}_1 + \dots + \lambda_n \mathbf{e}_n = \mathbf{0}$, i.e.

$$\lambda_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + \lambda_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

The last vector equation implies that $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$.

Our First Vector-Based Operation: The Scalar Product

Function $\cdot : \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R}, (x, y) \mapsto x \cdot y = \sum_{i=1}^n x_i y_i$

- Reduces vectors to a scalar (one-dimensional) number
 - Also called **dot product**, also notated $\langle x, y \rangle$ or $x'y$
 - $x \cdot x = \sum_{i=1}^n x_i^2$ ("sum of squares")
 - Practical usecase: checking orthogonality
 - ...of vectors $x, y : x \cdot y = 0$
 - ...of lines f, g in $\mathbb{R}^n$: $(x_f^1 - x_f^2) \cdot (x_g^1 - x_g^2) = 0$
 where x_f^1, x_f^2 are *distinct* points on f and x_g^1, x_g^2 are *distinct* points on g
- $\Rightarrow$ underlying reason: For x, y with angle θ it holds: $x \cdot y \propto \cos \theta$

Norms – an Introduction to Distances

- Measure distance between to objects
⇒ e.g. two points, two vectors, vector from origin etc.
- Universal (intuitive) criteria
 - Non-negative, and zero only if we don't have to move
 - Symmetric: same distance from A to B and from B to A
 - Detours increase the distance

Metric

For $\mathbb{X} = (X, +, \cdot)$, a function $d : X \times X \mapsto \mathbb{R}$ defines a **metric** on X if

1. non-negativity: $\forall x, y \in X : d(x, y) \geq 0$, and $d(x, y) = 0 \Leftrightarrow x = y$
2. symmetry: $\forall x, y \in X : d(x, y) = d(y, x)$
3. triangle inequality: $\forall x, y, z \in X : d(x, y) \leq d(x, z) + d(z, y)$

Norms – Normed Vector Spaces

- Interested in particular distances: lengths of vectors

Norm

For $\mathbb{X} = (X, +, \cdot)$, a **function** $\|\cdot\| : X \mapsto \mathbb{R}$ defines a **norm** on X if

1. non-negativity: $\forall x \in X : \|x\| \geq 0$, and $\|x\| = 0 \Leftrightarrow x = \mathbf{0}$
2. absolute homogeneity: $\forall x \in X, \lambda \in \mathbb{R} : \|\lambda \cdot x\| = |\lambda| \cdot \|x\|$
3. triangle inequality: $\forall x, y \in X : \|x + y\| \leq \|x\| + \|y\|$

Then, $(X, \|\cdot\|)$ forms a **normed vector space**.

- Examples: **p-Norm** (in $\mathbb{R}^n$): $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$

Max. norm (in $\mathbb{R}^n$): $\|x\|_\infty = \max_{i \in \{1 \dots n\}} |x_i|$

Natural Norm (in $\mathbb{R}$): $\|x\| = |x|$

Norms – Norm Induced Metrics

- **Norm-induced metric:** derive metric d_N from norm $\|\cdot\|_N \rightarrow d_N(x, y) := \|x - y\|_N$
 $\Rightarrow$ absolute homogeneity: $\forall x, y \in X \forall \lambda \in \mathbb{R} \ d_N(\lambda x, \lambda y) = |\lambda| d_N(x, y)$
 $\Rightarrow$ translation invariance: $\forall x, y, z \in X \ d_N(x + z, y + z) = d_N(x, y)$

- Continuity of the norm implies continuity of norm-induced metrics
 $\Rightarrow$ Continuity of the norm function $\|\cdot\| : X \mapsto \mathbb{R}, x \mapsto \|x\|$:

$$\forall \varepsilon > 0 \ \exists \delta > 0 : (\|x - y\| < \delta \Rightarrow \left| \|x\| - \|y\| \right| < \varepsilon)$$

$\Rightarrow$ Follows from inverse triangle inequality $\left| \|x\| - \|y\| \right| \leq \|x - y\|$

$\Rightarrow$ Intuition: points that are closer " $\|x - y\|$ " have similar magnitude " $\left| \|x\| - \|y\| \right|$ "

$\Rightarrow$ Sequence-continuity: we can pull in limits: $\lim_{n \rightarrow \infty} \|x_n\| = \left\| \lim_{n \rightarrow \infty} x_n \right\|$

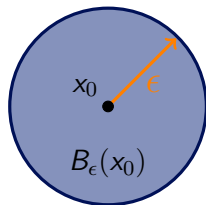
Norms – Euclidian Space

- Focus on Euclidean Vector Spaces $\mathbb{R}^n$ typically considered in Economics
 - **Euclidean Norm** (p-Norm for $p = 2$) as standard norm: $\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$
 $\Rightarrow$ norm-induced metric space $(\mathbb{R}^n, d_2)$ with

$$d_2(x, y) = \|x - y\|_2 = \left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{1/2} = \sqrt{(x - y) \cdot (x - y)}$$

- Crucial importance in Econometrics: Least Squares estimators
- Geometric intuition: direct distance (cf. Pythagorean Theorem)
- “The distance” usually refers to the Euclidean norm(-induced metric)

Sets and Balls using Metrics



- **Ball** of radius $\epsilon > 0$ around x_0 :
all points with distance to x_0 "smaller" than ϵ
 - strictly ($d(x, x_0) < \epsilon$): **open** ball $B_\epsilon(x_0)$
 - weakly ($d(x, x_0) \leq \epsilon$): **closed** ball $\bar{B}_\epsilon(x_0)$
 - "closed balls include the **boundary**, open balls do not"

- Two types of points: interior and boundary points ($\text{int}(A)$ vs. ∂A)
- Open set: only interior, no boundary points: $A = \text{int}(A)$
- Closed set: also includes all boundary points: $A = \text{int}(A) \cup \partial A$
- Bounded set: bounded distance of elements: $\exists x \in X \exists r < \infty : A \subseteq B_r(x)$
- Compact set $\iff$ closed and bounded [Heine-Borel Theorem]
 $\Rightarrow$ fundamentally important for optimization (together with continuity)

Sets and Balls using Metrics – Formal Definitions

- ε -open Ball around x_0 :

$$B_\epsilon(x_0) = \{x \in X : d(x, x_0) < \epsilon\} \quad [\text{Generally}]$$

$$B_\epsilon(x_0) = \{x \in X : \|x - x_0\| < \epsilon\} \quad [\text{Norm-induced metric}]$$

- Interior point:

$$x \in \text{int}(A) \Leftrightarrow \exists \epsilon > 0 : B_\epsilon(x) \subseteq A$$

- Boundary point:

$$x \in \partial A \Leftrightarrow \forall \epsilon > 0 : B_\epsilon(x) \cap A \neq \emptyset \wedge B_\epsilon(x) \setminus A \neq \emptyset$$

Helpful Theorems 1/3

Properties of Open and Closed Sets

Consider a metric space $(\mathbb{X}, d)$. Then,

(o.i) $\emptyset$ and X are open in $\mathbb{X}$.

(o.ii) A set $A \subseteq X$ is open if and only if its complement $A^c = X \setminus A$ is closed.

(o.iii) The union of an arbitrary (possibly infinite) collection of open sets is open.

(o.iv) The intersection of a finite collection of open sets is open.

(c.i) $\emptyset$ and X are closed in $\mathbb{X}$.

(c.ii) A set $A \subseteq X$ is closed if and only if its complement $A^c = X \setminus A$ is open.

(c.iii) The union of a finite collection of closed sets is closed.

(c.iv) The intersection of a (possibly infinite) collection of closed sets is closed.

Take-away: check complements and/or decompose into $\cup/\cap$ of simple sets!

Helpful Theorems 2/3

Closedness and Sequences

Suppose that $\mathbb{X} = (X, +, \cdot)$ is a real vector space, and let $B \subseteq X$. Then, B is closed if and only if, for any convergent sequence $\{x_n\}_{n \in \mathbb{N}}$ over B , i.e. $\forall n \in \mathbb{N} : x_n \in B$, it holds that $\lim_{n \rightarrow \infty} x_n \in B$.

Checking Boundedness

Suppose $(\mathbb{X}, d)$ is a metric space where d is norm-induced, i.e. for $x, y \in X$, $d(x, y) = \|x - y\|$. Let $A \subseteq X$. Then, A is bounded if the norm is bounded on A , i.e. $\exists b < \infty : (\forall x \in A : \|x\| < b)$.

Helpful Theorems 3/3

Weak Inequalities and the Limit: Functions

Suppose that $\mathbb{X} = (X, +, \cdot)$ is a real vector space, $f: X \mapsto \mathbb{R}$ and $g: X \mapsto \mathbb{R}$ so that $\forall x \in X: f(x) \leq g(x)$ (in function notation: $f \leq g$). Let $x_0 \in X$, and suppose that $\exists f_0, g_0 \in \mathbb{R}$ so that $\lim_{x \rightarrow x_0} f(x) = f_0$, $\lim_{x \rightarrow x_0} g(x) = g_0$. Then, it holds that $f_0 \leq g_0$.

Weak Inequalities and the Limit: Sequences

Suppose that $\mathbb{X} = (X, +, \cdot)$ is a real vector space. Let $\{x_n\}_{n \in \mathbb{N}}$ and $\{y_n\}_{n \in \mathbb{N}}$ be convergent sequences over X , i.e. $\forall n \in \mathbb{N}: x_n, y_n \in B$, with limits $x \in X$ and $y \in X$, respectively. If $\forall n \in \mathbb{N}$, it holds that $x_n \leq y_n$, then, we also have $x \leq y$.

Convergence in Normed Spaces – Sequences

- Recall $\mathbb{R}$: real sequence $\{x_n\}_{n \in \mathbb{N}}$ is convergent with limit $x \in \mathbb{R}$ if

$$\forall \epsilon > 0 \exists N \in \mathbb{N} : (\forall n \in \mathbb{N}, n \geq N : |x_n - x| < \epsilon)$$

- Recall: $|\cdot|$ is the **natural norm** of $\mathbb{R}$, so that equivalently

$$\forall \epsilon > 0 \exists N \in \mathbb{N} : (\forall n \in \mathbb{N}, n \geq N : \|x_n - x\| < \epsilon)$$

- Extends to **normed VS** $(X, \|\cdot\|_X)$: $\{x_n\}_{n \in \mathbb{N}}$ where $\forall n \in \mathbb{N} : x_n \in X$ is convergent with limit $x \in X$ if

$$\forall \epsilon > 0 \exists N \in \mathbb{N} : (\forall n \in \mathbb{N}, n \geq N : \|x_n - x\|_X < \epsilon)$$

- For sequences over $\mathbb{R}^n$:

$$\lim_{n \rightarrow \infty} (x_1, \dots, x_n) = (\lim_{n \rightarrow \infty} x_1, \dots, \lim_{n \rightarrow \infty} x_n)$$

Convergence in Normed Spaces – Functions

- Recall: for a univariate, real-valued function, i.e. $f: X \mapsto Y$ with $X, Y \subseteq \mathbb{R}$, $f_a \in Y$ is the limit of f at $a \in X$ if

$$\forall \epsilon > 0 \exists \delta > 0 : (\forall x \in X : |x - a| < \delta \Rightarrow |f(x) - f_a| < \epsilon)$$

- General function $f: X \mapsto Y$ where $X \subseteq (\mathbb{X}, \|\cdot\|_X)$, $Y \subseteq (\mathbb{Y}, \|\cdot\|_Y)$:

$$\forall \epsilon > 0 \exists \delta > 0 : (\forall x \in X : \|x - a\|_X < \delta \Rightarrow \|f(x) - f_a\|_Y < \epsilon)$$

$\Rightarrow$ Can equivalently write $x \in B_\delta(a)$ for $\|x - a\|_X < \delta$

- More general definitions for any metric space (not “just” norm-induced) exist, less relevant to us

Convergence in Normed Spaces – Continuity

- Recall: continuity for a univariate, real-valued function $f(a) = \lim_{x \rightarrow a} f(x)$

$$\forall \epsilon > 0 \exists \delta > 0 : (\forall x \in B_\delta(x_0) : |f(x) - f(x_0)| < \epsilon)$$

- General function $f: X \mapsto Y$ where $X \subseteq (\mathbb{X}, \|\cdot\|_X)$, $Y \subseteq (\mathbb{Y}, \|\cdot\|_Y)$:

$$\forall \epsilon > 0 \exists \delta > 0 : (\forall x \in B_\delta(x_0) : \|f(x) - f(x_0)\|_Y < \epsilon)$$

- For continuous functions limits can be pulled in:

$$\lim_{x \rightarrow x_0} f(x) = f(\lim_{x \rightarrow x_0} x) = f(x_0)$$

- Disprove continuity:

$$\text{find } x_n \xrightarrow{n \rightarrow \infty} x_0 \text{ with } f(x_n) \not\xrightarrow{n \rightarrow \infty} f(x_0) \text{ (non-existent or different limit)}$$

Convexity

- Often domains aren't VS $\Rightarrow$ use sets that focus on special linear combinations

Convex Combination, Convex Set

Suppose $\mathbb{X}$ is a real vector space. A **convex combination** x^c of the vectors $x_1, \dots, x_k \in X$ is a LC $x^c = \sum_{i=1}^k \lambda_i x_i$, for which $\forall i \in \{1, \dots, k\} : \lambda_i \geq 0$ and $\sum_{i=1}^k \lambda_i = 1$.

A set $A \subseteq X$ is **convex** if it contains all convex combinations of any two of its elements, i.e. $\forall a_1, a_2 \in A \quad \forall \lambda \in [0, 1] : \lambda a_1 + (1 - \lambda) a_2 \in A$.

- Intuition: convex combination of 2 vectors: $\{\lambda x + (1 - \lambda)y : \lambda \in [0, 1]\}$
 $\Rightarrow$ **line** connecting x and $y \rightarrow$ the larger λ , the more we move from y to x
- Graphical test in $\mathbb{R}^2$ and $\mathbb{R}^3$: connecting lines fully contained in set?

Convexity-preserving Operations

Convexity-preserving Operations

Suppose $\mathbb{X} = (X, +, \cdot)$ is a real vector space. Then,

1. $\emptyset$ and X are convex.
2. if $A \subseteq X$ is convex, then so is $\alpha A := \{\alpha \cdot a : a \in A\}$ for any $\alpha \in \mathbb{R}$.
3. if $A, B \subseteq X$ are convex, then so is $A + B := \{a + b : a \in A, b \in B\}$.
4. if $\{A_i\}_{i \in I}$ is a (possibly infinite) collection of convex sets, then $\bigcap_i A_i$ is convex.

- Can be used to simplify analysis of sets

In-Class Exercises I

Question 1 If $x = (1, 2, 4)'$ and $y = (3, 0, 2)'$, what is $(2x) \cdot y$?

Question 2 Consider $x = (1, 2, 0)'$ and $y = (2, 4, 1)'$. Are they linearly independent? What is their span? Are additional vectors required to span $\mathbb{R}^3$?

Question 3 Draw the French metro distance d_F (defined below). Is it a metric?

$$d_F(x, y) = \|x - y\| \quad \text{if } \exists \alpha \in \mathbb{R} : x = \alpha y \quad \text{or} \quad d_F(x, y) = \|x\| + \|y\| \quad \text{otherwise}$$

Question 4 Which of the following sets are convex?

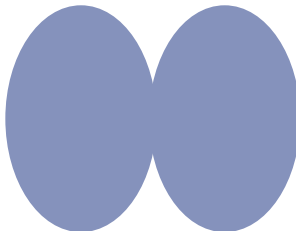
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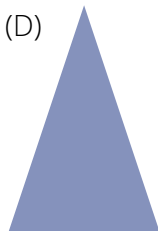
(B)



(C)



(D)



Recap Chapter 1

Vector Spaces Generalization of linear algebra to broader objects

- Linear Combinations and bases form the VS building blocks
- Norms and metrics structure the objects

Limits and Continuity Translating definitions to Vector Spaces

- A more general distance concept extends the univariate definitions
- Fundamental concepts remain unchanged

Set Properties

- Adjusted definitions for open/closed, compact, convex sets
- Relevance of "regular" sets for handling and optimization

That's all Folks!

Please take a look at the problem set that we will discuss tomorrow morning.

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